# Probability Density Function and Probability Mass Function

Two coin tosses
X= Number of Heads
Possible values : X = 0,1,2
HH(2),HT(1),TH(1),TT(0)

We know X can be 0, 1 or 2 . But how likely is each value?
**Probability Distribution**

Discrete Random Variable--> PMF

**1. PROBABILITY MASS FUNCTION (PMF)**

PMF tells us the probability associated with each possible value of discrete random variable.

Example:
X= number of heads in two coin tosses
Possible values : 0,1,2
Probability table

| X | P(X) |
|---|---|
| 0 | 0.25 |
| 1 | 0.5 |
| 2 | 0.25 |

Probability of 0 head = 0.25
Probability of 1 head = 0.50
Probability of 2 heads = 0.25

**Technically**

For a discrete random variable X, the PMF is

$$p(x) = P(X = x)$$

It gives the probability that the random variable takes a specific value x.

**Two important conditions**

$$P(X = x) >= 0$$

and

$$\sum P(X = x) = 1$$

If we consider every possible outcome, their probabilities must add up to 1(e

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**Example: Two coin Tosses**

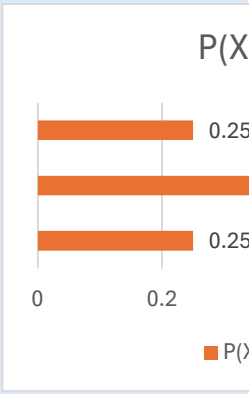
Sample Space

S = {HH,HT,TH,TT} n(S) = 4

Define

X = Number of Heads

| Outcome | X | Probability |
|---|---|---|
| HH | 2 | "1/4" |
| HT | 1 | "1/4" |
| TH | 1 | "1/4" |
| TT | 0 | "1/4" |




Cobining the identical X values

| X | P(X) | P(X) |
|---|---|---|
| 0 | 0.25 | "1/4" |
| 1 | 0.5 | "2/4" |
| 2 | 0.25 | "1/4" |

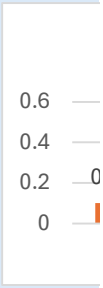
Therefore:

**0.25+0.50+0.25 = 1**

**Business Example: Number of Purchases**

Suppose an online store studies how many products customers purchase per order.

| Products Purchase | | Probability |
|---|---|---|
| 0 | | 0.1 |
| 1 | | 0.45 |
| 2 | | 0.3 |
| 3 | | 0.15 |




Ask:

What is the probability that a customer buys exactly 2 products?

**P(X = 2)= 0.30**

What is probability that a customer buys at least 2 products?

**P(X >=2)**

0.45

What is the probability that a customer buys at most 2 products?

**P(X<=2)**

0.85

# Probability Density Function and Probability Mass Function

**A random variable X has the following Distribution**

| X | P(X) |
|---|---|
| 1 | 0.2 |
| 2 | 0.3 |
| 3 | 0.4 |
| 4 | 0.1 |

**Questions:**

1 Is this a valid PMF? 1 Therfore Valid

2. What is Probability of exactly 3 P(X= 3)? P(X= 3) = 0.40

3. What is the probability of at least 3 P(X >=3) P(X>=3) = 0.40+0.10

**PROBABILITY DENSITY FUNCTION (PDF)**

PMF works for discrete random variables. What is the variable is continuous?

Example:

Customer Waiting Time

In this case we will use probability density function

**PDF describes how probabilty is distributed across the possible values of a continuous random variable**

PMF--> Probability at discrete value

PMF--< Probability is obtained from an area under the curve

**Technically**

For a continuous random variable X, a probability density function f(x) satisfies:

$$f(x) \geq 0$$

and the total area under the curve is

$$\int f(x)dx = 1$$

the range for integration lies from $-\infty$ to $\infty$

Probability between a and b

$$P(a \leq X \leq b) = \int f(x)dx$$

**Area under the curve = Probability**

For a continuous random variable:

$$P(X = x) = 0$$

Why?

There are infinitely many possible values, so probability is associated with intervals, not an individual exact point.

For example:

Instead of asking

Probability that the delivery time is exactly 5.0000000 minutes?

# Probability Density Function and Probability Mass Function

Normally
Probability that delivery time is between 5 and 10 minutes.

**Visual Example: Delivery time**
Suppose

X = delivery time

5 minutes and 10 mintutes
P(5<=X<=10) it is represented by the shaded region.

**Business Use case: -- Logistics**
What proportion of deliveries take between 30 and 45 minutes?
**Business Decisions:**
It will help with:
1. Delivery SLA analysis
2. Customer Expectations
3. Workforce planning
4. Route planning
5. Performance monitoring.

| PMF | PDF |
|---|---|
| Discrete random variable | Continuous random variable |
| Probability of a specific value | Probability over an interval |
| Uses individual probability values | Uses area under curve |
| Sum of probabilities = 1 | Total area = 1 |
| Bar/point representation | Smooth density curve |
| Number of customers | Delivery time |
| Number of defects | Product weight |

**EXPECTED VALUE**
Expected value is the probability-weighted average of all possible outcomes of a random variable.

**Expected value tells us the long-run average outcome we would expect if the experiment were repeated many times.**

**Technically**
For a discrete random variabel:
$E(X) = \sum xP(X= x)$

x= possible value
P(X=x) = probability of that value
xP(X=x)= weighted contribution
Sum of all contributions

# Probability Density Function and Probability Mass Function

**Example: Fair Die**

Possible outcomes:

{1,2,3,4,5,6}

Each has:

P(X=x)= 1/6

Therefore

E(X) = 1(1/6)+2(1/6)+3(1/6)+4(1/6)+5(1/6)+6(1/6)

3.5

Can a die actually show 3.5?

No

Therefore:

Expected value does not necessarily have to be one of the possible outcomes

It represents the long-run average.

**Business Example: Expected Daily Profit**

| Profit | Probability |
|---|---|
| 1000 | 0.2 |
| 2000 | 0.5 |
| 5000 | 0.3 |

Calculate E(X) SUMPRODUCT 2700

Step 1:

1000(0.20) =200

Step2:

2000(0.50)=1000

Step3:

5000(0.30) = 1500

Step 4 :

E(X) = 200+1000+1500

E(X) = 2700

**Interpretation:**

**If the probability model remains valid over multiple repetitions, the long run average profit will be approximately 2700**

**Interview Questions:**

What is PMF?

What is PDF?

What is difference between PMF and PDF?

What must the probabilities in a PMF add up to?

What does the area under the curve represent?

What is Expected value?

Expected value is the probability-weighted average of all possible outcomes of a random variable.

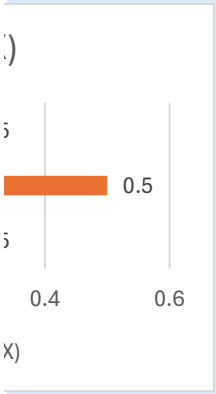
**Can expected value be a value that never actually occurs?**

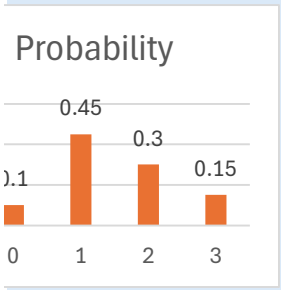
Yes, A die has expected value 3.5, but it cannot appear on the die.

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quals to 1).

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